Assignment 2

1.RNN vs Transformer (Differences, Advantages & Limitations)

->Recurrent Neural Network (RNN) is a type of neural network designed to process sequential data. It **reads information one step at a time** and tries to remember previous information while moving forward.

Example: **Sentence->** I love studying Artificial Intelligence.

RNN reads:

I
↓
love
↓
studying
↓
Artificial
↓
Intelligence

one word at a time.

Advantages of RNN:

- Good for sequential data.
- Simple architecture.
- Can remember previous information to some extent.

Limitations of RNN:

- Processes data slowly because it reads one word at a time.
- May forget important information in long sentences.
- Difficult to train on very large datasets.

->**Transformer** is a neural network architecture that processes all words in a sentence simultaneously and uses an **Attention Mechanism** to understand relationships between words.

Example: Sentence:->Yugaant studies hard because he wants good marks.

The Transformer understands that:

he = Yugaant

by analyzing the entire sentence at once.

Advantages of Transformer:

- Processes data much faster.
- Handles long sentences effectively.
- Better at understanding relationships between words.
- Can be trained on massive datasets.
- Used in modern AI systems like ChatGPT and Gemini.

Limitations of Transformer:

- Requires high computing power.
- Needs large amounts of training data.
- More expensive to train compared to RNN.

Key Difference:

RNN	Transformer
Reads one word at a time	Reads the entire sentence at once
Slower processing	Faster processing
May forget earlier information	Better memory of context
Suitable for smaller tasks	Suitable for large AI models
Used in older AI systems	Used in modern AI systems

Attention Mechanism:

->**Attention Mechanism** helps the Transformer identify important words and understand how different words in a sentence are related to each other.

Example

The animal didn't cross the road because it was too tired.

The Transformer understands that:

it → animal

by paying attention to the relationship between the words.

2.How Transformers Revolutionized AI

Before Transformers:

AI mainly used:

- RNN (Recurrent neural network)
- LSTM (Long short-term memory)

Problems:

❌ Slow training

❌ Forget long context

❌ Difficult to scale

-> In 2017 researchers at Google introduced the Transformer architecture in the paper: Attention Is All You Need

What Changed?

1. Faster Training:

RNN:

Word1
↓
Word2
↓
Word3

Transformer:

Word1
Word2
Word3

All processed together.

2. Better Understanding:

->Transformer can connect: Yugaant studies hard because he wants good marks.

and understand:

->he = Yugaant, even if many words separate them.

3. Larger Models Became Possible:

Companies could train:

Millions of parameters
↓
Billions
↓
Hundreds of billions

which was very difficult before.

4. Birth of Modern AI:

Transformers enabled:

- ChatGPT
- Gemini
- Claude
- DeepSeek

3.What are LLM's?

->**Large Language Models (LLMs)** are specialized Deep Learning models designed to understand and generate human language. They are built using **Transformer architecture** and trained on massive amounts of text data using **Self-Supervised Learning**.

->LLMs can understand natural language inputs such as questions, commands, conversations, and documents. They analyze the relationships between words, grammar, context, and meaning to generate human-like responses.

->Unlike general Deep Learning models that may focus on images, videos, or audio, LLMs are specifically designed for language-related tasks such as:

- Question Answering
- Conversation
- Essay Writing
- Summarization
- Translation

- Code Generation
- Content Creation

->During training, LLMs learn patterns from billions of words and store this knowledge in the form of **parameters**. When a user asks a question, the model converts the text into tokens, analyzes the context using the Transformer architecture, and predicts the most appropriate next words to generate a response.

Example:

Suppose a user asks:

What is Artificial Intelligence?

The LLM:

1. Convert the sentence into tokens.
2. Understands that the user is asking for a definition.
3. Uses its learned knowledge (parameters).
4. Generates a response word by word.

Output:

Artificial Intelligence is a branch of computer science that enables machines to perform tasks that normally require human intelligence.

Real-World Examples of LLMs

- ChatGPT
- Gemini
- Claude
- DeepSeek
- Llama

4. How LLM's Generate Text?

->LLMs generate text by converting user input into tokens, analyzing context using Transformer architecture, and predicting the most probable next tokens using learned parameters until a complete response is produced.

->It **hallucinates** (It gives a convincing answer more than a correct answer)

How it works?

User Question

↓

Convert into Tokens

↓

Transformer understands Context

↓

Uses Parameters (Learned Knowledge)

↓

Predicts Next Token

↓

Generates Response

Example:

User Input:

What is Artificial Intelligence?

<u>Step 1: Tokenization:</u>

->The sentence is converted into tokens.

<u>Ex:</u> What | is | Artificial | Intelligence |?

<u>Step 2: Context Understanding:</u>

->The Transformer understands that the user is asking for a definition.

<u>Step 3: Parameter Usage:</u>

->The model uses its learned parameters and language patterns related to Artificial Intelligence.

<u>Step 4: Next Token Prediction:</u>

The model predicts:

Artificial
↓
Intelligence
↓
is
↓
a
↓

branch
↓
of
↓
computer
↓
science
...

one token at a time.

Step 5: Final Response:

->Artificial Intelligence is a branch of computer science that enables machines to perform tasks that normally require human intelligence.

5. Parameters:

->Parameters are the numerical values inside a neural network that store the knowledge learned during training.

Think of parameters as:

AI Knowledge = Parameters

Why Are They Numbers?

->Computer understand numbers

->During training the model adjusts billions of numbers

->Parameters are stored in permanent storage.

Why Are More Parameters Important?

More Parameters

↓

More Learning Capacity

↓

Better Understanding

Example:

->During training, the model sees:

The sky is blue. (many times)

->Eventually, the parameters learn:

"The sky is" → "blue" is highly likely.

->**The sentence itself is not stored. The learned pattern is stored in parameters.**

6. Tokens

->Tokens are small pieces of text that an LLM uses to understand and generate language. Before processing a sentence, the model converts the text into tokens. A token can be a complete word, part of a word, punctuation mark, number, symbol, or sometimes spaces. The way text is divided into tokens depends on the tokenizer used by the model.

Example:

<u>User Input:</u>

What is Artificial Intelligence?

<u>Possible Tokens:</u>

What
is
Artificial
Intelligence
?

->The LLM processes these tokens instead of the original sentence.

<u>Another example:</u>

unbelievable

<u>Possible Tokens:</u>

un
believ
able

->depending on the tokenizer.

7. Context Window

->A Context Window is the amount of text (tokens) that an **LLM can see, remember, and use at one time while generating a response**. It **acts like the AI's temporary working memory, similar to RAM in a computer**. If the context window becomes **full, older tokens may be removed so that newer tokens can be processed.**

Example:

Conversation:

User: My name is Yugaant.
User: I study CSE AIML.
User: What is my name?

AI: Your name is Yugaant.

->The AI can answer correctly because the previous messages are still inside its context window.

->If the conversation becomes extremely long and the context window gets full:

Old Messages ❌
New Messages ✅

->The AI may forget earlier parts of the conversation because they are no longer inside the context window.

8. Open-Source VS Closed Source

->Open Source and Closed Source models differ in how much of the AI model is made available to the public. **Open-Source models allow developers to access and modify the model**, while **Closed Source models allow users to use the AI but do not provide access to its internal parameters, architecture, or implementation details.**

-> Companies choose between open and closed source based on their business goals. **Some companies keep models closed to protect their technology and earn revenue from providing access**, while **others make models open to encourage research, innovation, adoption, and community contributions.**

<u>Example:</u>

Open Source Model

->Llama (Large Language Model Meta AI) and DeepSeek are examples of open-source models.

Developers can:

->Download the model
->Run it on their own systems
->Modify it
->Fine-tune it for specific tasks

<u>Example:</u>

A hospital can download Llama and train it further on medical data to create a healthcare assistant.

Closed Source Model

ChatGPT, **Gemini**, and **Claude** are examples of closed-source models.

Users can:

->Use the AI, ask questions, Generate content

But they cannot:

->View the parameters
->Modify the model
->Access the internal architecture

<u>Example:</u>

You can use ChatGPT to ask questions, but you cannot download and modify the model yourself.

9. LLM Benchmarks and Evaluation

-> LLM Benchmarks are like exams for AI models. They are used to **compare different AI models and measure their capabilities** in tasks **such as logical reasoning, mathematics, coding, language understanding, summarization, and question answering**. The b<u>enchmark scores help determine which model performs better in different areas</u>.

Simple Example:

<u>Imagine:</u>

Students
↓
Math Exam
English Exam
Science Exam
↓
Marks

Similarly:

ChatGPT
Gemini
Claude
DeepSeek
↓
Reasoning Test
Math Test
Coding Test
Language Test
↓
Scores

->The model with higher scores generally performs better in those tasks.

10. Prompt Engineering

-> Prompt Engineering is the practice of creating clear and detailed instructions (prompts) to help an AI model generate accurate, relevant, and useful responses.

Example:

Simple Prompt:

Explain AI.

Better Prompt:

Explain AI to a beginner using simple language and real-world examples.

->The second prompt produces a more useful answer because it gives clear instructions.

Common Prompting Techniques:

1. Clear Instructions:

Instead of: Explain AI.

Use: Explain AI in simple language with 3 examples.

2. Specify the Role:

Example: Act as a Computer Science professor and explain Neural Networks.

->The AI responds like a teacher.

3. Give Context:

Example: I am a second-year CSE AIML student. Explain Transformers in beginner-friendly language.

->The AI gives an answer suitable for your level.

4. Specify Output Format:

Example: Explain AI in:
1. Definition
2. Example
3. Advantages
4. Applications

->The answer becomes organized.

11. AI Agent

An AI Agent is an AI system that can understand a goal, make decisions, plan steps, use tools, and perform actions to complete a task with minimal human guidance.

Example:

Task: Plan a trip to Chennai.

The AI Agent can:

Search flights
↓
Compare prices
↓
Find hotels
↓
Prepare travel plan

->instead of only giving information.

Difference Between LLM and AI Agent:

LLM:

Question
↓
Answer

Example:

What is the best hotel in Chennai?

Provides answer.

AI Agent:

Goal
↓
Search hotels
↓
Compare options
↓

Make recommendation-> Book hotel

->Can perform actions.

12. RAG (Retrieval Agumented Generation) Fundamentals

->RAG (Retrieval-Augmented Generation) is a technique that helps an LLM retrieve relevant information from external sources before generating a response.

-> Instead of relying only on its stored parameters, the model can access existing information from sources such as websites, databases, PDFs, or company documents and use that information to generate a more accurate answer.

Example:

Suppose ChatGPT was trained in 2024.

You ask:

Who won yesterday's IPL match?

->The information is not available in the model's parameters because the match happened after training.

RAG works as follows:

User Question
↓
Retrieve Relevant Information
(from news websites, databases, etc.)
↓
Provide Information to LLM
↓
LLM Generates Answer

->The AI does not create new knowledge or perform its own research. Instead, it retrieves existing information from external sources and uses it to generate a response.

13. Multimodal AI

-> Multimodal AI is an AI system that can understand and work with multiple types of data such as text, images, audio, video, and speech. Instead of processing only text, it can combine different forms of information to understand and generate responses.

Example:

Suppose you upload a photo of a plant and ask:

Which plant is this?

The AI:

Image
↓
Analyzes Plant
↓
Generates Text Answer

Here it uses:

Image + Text

->so, it is a Multimodal AI.

14. Model Fine-Tuning:

->Model Fine-Tuning is the process of training an already trained AI model with specialized data to make it better at a specific task or field.

Simple Example

General AI Model
↓
Medical Data
↓
Fine-Tuning
↓
Medical AI Assistant

Just like:

Computer Science Student
↓
Studies AI in Depth
↓
AI Engineer

->The student already had basic knowledge and learned more about a specific field. Similarly, a fine-tuned model uses its existing knowledge and becomes specialized in a particular domain.

Difference From RAG:

RAG

↓

Uses external information when needed

↓

Model does not change

Fine-Tuning

↓

Learns from specialized data

↓

Model parameters change

15.Why Enterprises Use Multiple Models:

->Enterprises use multiple AI models because different models are specialized for different tasks. Using multiple models helps organizations improve performance, reduce cost, increase reliability, and choose the best model for each requirement.

Example:

Customer Support → ChatGPT

Image Analysis → Gemini

Coding Tasks → Coding Model

->The company uses different models because each model performs better in certain areas.

Why Not Use One Model?

Suppose a company needs: Customer Support, Code Generation, Image Analysis Document, Summarization, Speech Recognition

->One model may be excellent at coding but average at image understanding.

->Another model may be excellent at image analysis but expensive for simple tasks.

->So, companies use multiple models.

Main Reasons Enterprises Use Multiple Models

1. Different Strengths:

Model A → Better at Coding

Model B → Better at Images

Model C → Better at Reasoning

2. Lower Cost:

Simple tasks: Use smaller cheaper model

Complex tasks: Use powerful model

3. Better Performance: Use the best model for each task.

4. Reliability: If one model fails, Switch to another model.

16.AI Safety, Privacy & Hallucinations:

1.AI Safety:

->AI Safety ensures that AI systems operate correctly, reliably, and without causing harm to people or society.

Example:

Self-driving car:

Camera sees Red Light
↓
AI decides to Stop

If AI misunderstands:

Red Light → Green Light

->it can cause an accident.

->That's why AI Safety is important.

2.AI Privacy:

->AI Privacy is the protection of personal, confidential, and sensitive information used by AI systems.

Why is Privacy Needed?

->Users may share: Passwords, Bank Details, Medical Records, Personal Information

->AI systems must protect this data.

Example:

Hospital AI:

Patient Records
↓
AI System

->The records should not be visible to unauthorized people.

->Privacy ensures user data remains secure.

3. AI Hallucinations:

->AI Hallucination occurs when an AI generates information that sounds correct and convincing but is actually incorrect, misleading, or completely made up.

Why Does Hallucination Happen?

You learned:

LLM
↓
Predicts Next Token

->The model does not truly know facts.

It predicts:

Which word is most likely next?

Sometimes the prediction is wrong.

Then hallucination occurs.

Example:

User asks:

Who invented the smartphone in 1890?

->There was no smartphones in 1890.

But AI may still confidently generate:

John Smith invented the smartphone in 1890.

->This sounds believable.

->But it is false, this is hallucination.

17.Comparative Analysis Of Leading AI Models:

->Comparative Analysis of Leading AI Models is the process of comparing different AI models based on their strengths, capabilities, performance, and use cases to determine which model is best suited for a particular task.

Example:

->Different AI models are strong in different areas:

ChatGPT → Reasoning, Coding, Conversation

Claude → Long Documents, Writing, Summarization

Gemini → Images, Videos, Multimodal Tasks

DeepSeek → Coding, Open-Source Development

Llama → Fine-Tuning, Private Enterprise AI

->Just like different specialist doctors handle different medical problems, different AI models excel at different tasks. Companies compare these models and choose the one that best fits their requirements.